

Appendix A. Stochastic conformance checking with guarantees – Proofs

This document contains formal proofs that justify the proposed stochastic conformance checking approach.

Proposition 1 (Well-Formedness). *The intersection of two well-formed FLMCs is a well-formed FLMC.*

Proof. By construction, the intersection model contains only reachable states, as these correspond to pairs of reachable states of the original models. Dead states are removed according to the definition of intersection. The resulting model is also split by construction, since transitions on different symbols lead to different state pairs, as they lead to different states in the original models. Finally, the empty sequence is not accepted by the intersection model, since it is not accepted by either of the original models. $\square$